

Orbital magnetic susceptibility and de Haas–van Alphen effect of a flat band from quantum geometry

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A band that is exactly flat at zero magnetic field has zero velocity, a divergent effective mass, and no Fermi surface, so the standard theory of orbital magnetism does not apply. We show that the magnetic field generates an effective dispersion. In the expansion organized by the Moyal algebra of the noncommutative kinetic momenta, the band energy acquires a first-order shift $B\mathcal{M}(\mathbf{k})$, set by the orbital moment of the Bloch state, and a second-order shift $\frac{1}{2}\mathcal{X}(\mathbf{k})B^2$, whose gauge-invariant coefficient we obtain in closed form. These two functions, together with the Berry curvature Ω , fix the entire orbital response. The susceptibility is purely geometric, and for a generic isolated flat band it reproduces the established gauge-invariant response formula. With the chemical potential pinned to the flat band, the $\mathcal{M} = 0$ contour becomes an emergent Fermi surface, and bosonizing excitations in its vicinity yields a modified Lifshitz–Kosevich formula with level spacing quadratic in B . The phase shift of the oscillations is given jointly by the Berry phase around the emergent Fermi surface and $\mathcal{X}$. When the band is narrow rather than exactly flat, the oscillation frequency crosses over, with a parameter-free profile, from the zero-field Fermi-surface area to the area of the $\mathcal{M} = 0$ contour. We verify our analytical findings numerically by exactly diagonalizing a flattened Chern lattice model.

I. INTRODUCTION

The orbital magnetic response of a metal is organized by its Fermi surface. The Landau–Peierls susceptibility is built from band curvature at the Fermi level [1], and the de Haas–van Alphen (dHvA) effect measures extremal Fermi-surface areas through the Onsager relation [2], with amplitudes given by Lifshitz–Kosevich (LK) theory [3, 4]. An exactly flat band has none of these ingredients. The effective mass diverges, the group velocity vanishes, and at any partial filling there is no Fermi contour to quantize. Such bands are now routinely engineered, in twisted bilayer graphene near the magic angle [5–7], in line-graph and kagome lattices [8], and in further moiré materials, and in them only the wavefunctions depend on momentum. Their quantum geometry, described by the Berry curvature and the quantum metric, controls the superfluid weight [9, 10] and the structure of the Landau levels [8, 11].

For an isolated flat band, Ref. [12] showed that the Landau levels are those of the effective dispersion $B\mathcal{M}(\mathbf{k})$ set by the orbital moment of the Bloch states, with level spacing quadratic in B . For a special class of exactly flat Chern bands with ideal quantum geometry, Ref. [13] obtained closed-form Landau levels at all field strengths, and from them the quantum-oscillation spectrum. For a generic isolated flat band, what these levels imply for the dHvA effect, its frequency, amplitudes, and phase shift, remains an open question, and first order in B is not enough to answer it. The phase of the oscillations is an orbit area divided by B . A second-order term of the dispersion shifts the area at first order and so enters the phase at order one. There is also no guarantee that the semiclassical rules for a metal, Onsager quantization

with its Berry-phase and Roth corrections [14, 15], carry over to a dispersion the field itself generates. In parallel, the orbital magnetism of Bloch electrons, a classic subject [16], has come under intense investigation from the perspectives of quantum geometry [17–20] and of electron interactions [21–23]. These works compute the magnetization and the susceptibility, the first two derivatives of the free energy in the field. The dHvA oscillations are invisible to this expansion. At fixed temperature they vanish faster than any power of B as $B \rightarrow 0$, contributing to no order in the field. A unified theory that produces the smooth derivatives and the oscillations from the same starting point has been missing. Given only the zero-field wavefunctions of an isolated flat band, what is the full orbital response, smooth and oscillatory?

In this paper we derive this theory to second order in the field. At first order, each momentum state of the flat band acquires the energy $B\mathcal{M}(\mathbf{k})$ of its own orbital moment. At second order a coefficient $\mathcal{X}(\mathbf{k})$ enters, for which we derive a closed form from a systematic Moyal expansion of the magnetic Bloch problem. The field-generated band

$$\varepsilon(\mathbf{k}; B) = \varepsilon_0 + B\mathcal{M}(\mathbf{k}) + \frac{1}{2}B^2\mathcal{X}(\mathbf{k}) \quad (1)$$

then feeds the standard thermodynamic machinery, with one geometric addition: the phase-space measure acquires the Berry factor $1 + B\Omega$ [17]. The smooth response and the oscillations both follow from this band. All results below are expressed in these three functions of $\mathbf{k}$, evaluated on the zero-field wavefunctions.

The susceptibility is a purely geometric functional of $\mathcal{M}$, $\mathcal{X}$, and Ω , and for a generic isolated flat band we prove that it agrees with the exact resolvent formula of

Ref. [18]. With the chemical potential pinned to the flat band, the $\mathcal{M} = 0$ contour becomes an emergent Fermi surface, and bosonizing the excitations in its vicinity along the lines of Ref. [24] produces a modified Lifshitz–Kosevich series whose quantization phase carries an $O(1)$ contribution from $\mathcal{X}$ alongside the Berry phase. On the hole pocket of the lattice model of Sec. V, we find that the contribution from $\mathcal{X}$ is larger than the Berry phase. The form of the series is obtained without invoking the semiclassical quantization rule of cyclotron orbits. When the band is narrow rather than exactly flat, Landau quantization proceeds on contours of the composite dispersion, built-in plus field-generated, and the dHvA frequency crosses over from the zero-field Fermi-surface area to the area of the $\mathcal{M} = 0$ contour, with a parameter-free profile and a $1/B$ asymptotic behavior.

The expansion is controlled by the single parameter $x = B\xi^2$, with ξ the geometric length of the band. Each order of the Moyal expansion carries one factor of x , and the closed forms for $\mathcal{M}$ and $\mathcal{X}$ confirm the counting (Sec. IID). The derivation applies to an isolated, nondegenerate flat band whose moment $\mathcal{M}(\mathbf{k})$ is smooth with a single closed $\mathcal{M} = 0$ contour, with no condition on its quantum geometry. These assumptions are collected in Sec. IV where each is used. The scope is complementary to that of Ref. [13], which is exact in B but restricted to bands with ideal quantum geometry, while our expansion works to $O(B^2)$. Where the two approaches overlap, they agree: our expansion reproduces the weak-field sector of that solvable model.

The oscillatory part of the analysis uses coadjoint-orbit bosonization, an exact bosonic description of a Fermi surface in terms of its shape fluctuations [25]. It was applied to the de Haas–van Alphen effect of a Fermi gas and a Fermi liquid in Ref. [26], and extended in Ref. [24] to multiband systems, where the Berry curvature enters through the phase-space measure. We extend the construction of Ref. [24], which keeps the dispersion to first order in the field, to the full second-order dispersion. For a dispersive band the second-order term shifts the quantization phase only at $O(B)$. For a flat band, whose level spacing is quadratic in B , it enters at order one.

The paper is organized as follows. Section II constructs the field-generated dispersion and establishes its gauge structure. Section III derives the susceptibility. Its consistency with the established gauge-invariant response theory is proved for a generic flat band in Appendix A. Section IV derives the LK formula from bosonization. Section V checks the analytic results against exact diagonalization of a flattened Chern lattice model. Section VI extends the theory to narrow bands and derives the crossover of the dHvA frequency. Section VII concludes with the experimental outlook.

Notation. $\mathcal{M}(\mathbf{k})$ is the orbital moment of the Bloch state and $\mathcal{X}(\mathbf{k})$ the second-order dispersion coefficient; $\Omega(\mathbf{k})$ is the Berry curvature; M denotes the thermodynamic magnetization; Ξ is the grand potential; Δ is the full gap between the flat band and its nearest neighbor;

$\tilde{A}(\mathcal{M}')$ is the area enclosed by the moment contour $\mathcal{M}(\mathbf{k}) = \mathcal{M}'$, $A_{\mathcal{M}} = \tilde{A}(0)$ the area enclosed by the $\mathcal{M} = 0$ contour $\mathcal{C}_0$, and A_{occ} the area of the occupied region. Band indices appear as $b \neq v$ in sums over remote bands, where v labels the flat band and b the remote bands. We set $\hbar = e = 1$. Then B is the magnetic flux per unit cell, and the flux quantum is 2π .

II. THE FIELD-GENERATED DISPERSION

A. Moyal expansion

Take an isolated band that is exactly flat at $B = 0$, $\varepsilon_0(\boldsymbol{\pi}) = \varepsilon_0$, where $\boldsymbol{\pi}$ is the kinetic momentum: the argument of the zero-field Bloch Hamiltonian, promoted in a uniform field to the noncommuting pair $[\pi_x, \pi_y] = +iB$, and reducing to the crystal momentum as $B \rightarrow 0$. Switching on a perpendicular field generates a dispersion whose only possible origin is the momentum dependence of the wavefunctions. This section computes it to second order. Once both coefficients are in closed form, estimating their magnitudes by counting wavefunction derivatives and comparing successive orders yields the expansion parameter $x = B\xi^2$ (Sec. IID).

Because π_x and π_y do not commute, $\hat{H}(\boldsymbol{\pi})$ is a function of noncommuting arguments, and products of such functions are operator products. The standard remedy is to represent each operator by its Weyl symbol, an ordinary function of $\boldsymbol{\pi}$, with the operator product encoded in the Moyal product of the symbols. The magnetic Bloch problem for the periodic part of the wavefunction then becomes a star-eigenvalue problem [27, 28],

$$\hat{H} \star |v\rangle = |v\rangle \star \varepsilon, \quad \star = \exp \left[+ \frac{iB}{2} \epsilon_{ij} \overleftarrow{\partial}_i \overrightarrow{\partial}_j \right], \quad (2)$$

where $\hat{H}(\boldsymbol{\pi})$ is the zero-field Bloch Hamiltonian evaluated at the kinetic momentum, $\partial_i = \partial/\partial\pi_i$ here and throughout this section, and the arrows indicate which factor each derivative acts on: $\overleftarrow{\partial}_i$ differentiates the symbol on the left of the $\star$, $\overrightarrow{\partial}_j$ the symbol on the right. The star product organizes the field corrections order by order in B . We expand state and eigenvalue in powers of B ,

$$|v\rangle = |v^0\rangle + |v^1\rangle + \cdots, \quad \varepsilon = \varepsilon_0 + \varepsilon^{(1)} + \varepsilon^{(2)} + \cdots, \quad (3)$$

where $|v^n\rangle$ and $\varepsilon^{(n)}$ collect the $O(B^n)$ contributions, and projecting order by order gives them as explicit functionals of the zero-field band.

B. First order: the orbital moment

Since ε_0 is a constant, all its derivatives vanish and it star-multiplies as an ordinary number, so, using the normalization $\langle v | \star | v \rangle = 1$, the projected dispersion $\varepsilon =$

$\langle v|\star\hat{H}\star|v\rangle$ splits exactly into

$$\varepsilon = \varepsilon_0 \langle v|\star|v\rangle + \langle v|\star(\hat{H} - \varepsilon_0)\star|v\rangle. \quad (4)$$

The normalization $\langle v|\star|v\rangle = 1$ fixes the first term to the constant ε_0 . The field-generated dispersion comes entirely from the gap-weighted second term, at every order, since a dispersion cannot depend on the value of the constant ε_0 .

At $O(B)$ the cross terms $\langle v^0|(\hat{H} - \varepsilon_0)|v^1\rangle$ and its conjugate vanish, because $(\hat{H} - \varepsilon_0)$ annihilates $|v^0\rangle$ whether it acts to the right or to the left. What remain are the $\langle v^0|\dots|v^0\rangle$ terms, where each Moyal product in the second term contributes one derivative insertion $\frac{iB}{2}\epsilon_{ij}\partial_i\partial_j$. The insertion at the left Moyal product differentiates the product $(\hat{H} - \varepsilon_0)|v^0\rangle$ and splits in two by the product rule, giving three matrix elements in all. A useful relation, obtained by differentiating the zero-field eigenvalue equation $\hat{H}|v^0\rangle = \varepsilon_0|v^0\rangle$,

$$(\partial_i\hat{H})|v^0\rangle = -(\hat{H} - \varepsilon_0)|\partial_iv^0\rangle, \quad (5)$$

together with its Hermitian conjugate, converts the two matrix elements that carry $\partial\hat{H}$, and the dispersion reduces as follows:

$$\begin{aligned} \varepsilon^{(1)} &= \frac{iB}{2} \epsilon_{ij} \left[\langle \partial_iv^0|(\partial_j\hat{H})|v^0\rangle + \langle \partial_iv^0|(\hat{H} - \varepsilon_0)|\partial_jv^0\rangle \right. \\ &\quad \left. + \langle v^0|(\partial_i\hat{H})|\partial_jv^0\rangle \right] \\ &= \frac{iB}{2} \epsilon_{ij} (-1 + 1 - 1) \langle \partial_iv^0|(\hat{H} - \varepsilon_0)|\partial_jv^0\rangle \\ &= B\mathcal{M}(\boldsymbol{\pi}), \end{aligned} \quad (6)$$

where

$$\mathcal{M} \equiv -\frac{i}{2} \epsilon_{ij} \langle \partial_iv^0|(\hat{H} - \varepsilon_0)|\partial_jv^0\rangle. \quad (7)$$

Equation (7) is the orbital magnetic moment of the Bloch state [17, 20, 27, 28] in our sign conventions. It is gauge invariant as written. Under $|v^0\rangle \rightarrow e^{i\phi}|v^0\rangle$ the extra terms all carry $(\hat{H} - \varepsilon_0)|v^0\rangle$ or its conjugate, both zero.

To leading order, then, the field generates a dispersive band due to the momentum-dependent orbital moment, with bandwidth $(\mathcal{M}^{\max} - \mathcal{M}^{\min})B$ linear in B . Wherever $\mathcal{M}$ changes sign, the field-generated band straddles the flat-band energy, and the level set

$$\mathcal{C}_0 = \{\boldsymbol{\pi} : \mathcal{M}(\boldsymbol{\pi}) = 0\} \quad (8)$$

becomes a Fermi surface for a chemical potential pinned at ε_0 . Equation (8) is an emergent Fermi contour, present only at finite field, whose enclosed area is determined by the zero-field wavefunctions. A sign change of $\mathcal{M}$ is not guaranteed. A band whose moment has a definite sign, as symmetry may enforce, has no such contour, and the field then shifts the band without creating a Fermi surface. In this work we focus on the case where the contour exists.

C. Second order: the closed form of $\mathcal{X}$

At $O(B^2)$ the advantage of the split (4) remains. The raw expansion contains the unknown second-order state $|v^2\rangle$, but it enters the gap-weighted term only through $\langle v^0|(\hat{H} - \varepsilon_0)|v^2\rangle$ and its conjugate, both zero, so $|v^2\rangle$ drops out identically and never needs to be computed. The second term of Eq. (4) is expanded at $O(B^2)$ by distributing the derivative insertions of the two star products over the state expansion (3),

$$\begin{aligned} \varepsilon^{(2)} &= \left[\langle v^0|\star(\hat{H} - \varepsilon_0)\star|v^0\rangle \right]_{O(B^2)} \\ &\quad + 2 \operatorname{Re} \left[\langle v^0|\star(\hat{H} - \varepsilon_0)\star|v^1\rangle \right]_{O(B)} \\ &\quad + \langle v^1|(\hat{H} - \varepsilon_0)|v^1\rangle, \end{aligned} \quad (9)$$

where the subscript selects the stated order of the star insertions $\frac{iB}{2}\epsilon_{ij}\partial_i\partial_j$ and the $\langle v^0|\dots|v^2\rangle$ terms have already been dropped. The first bracket carries two insertions, both at the same star product, which contributes with weight $\frac{1}{2}(\frac{iB}{2})^2$ from the exponential, or one at each, with weight $(\frac{iB}{2})^2$. The product rule distributes the derivatives over $(\hat{H} - \varepsilon_0)$ and the states, and the identity (5) removes the terms in which no derivative acts on $|v^0\rangle$. The second bracket carries one insertion. Its surviving terms reduce under Eq. (5) to a single Im matrix element. Carrying out this bookkeeping leaves

$$\begin{aligned} \varepsilon^{(2)} &= -\frac{B^2}{8} \epsilon_{ij}\epsilon_{kl} \langle v^0|(\partial_k\partial_i\hat{H})|\partial_j\partial_lv^0\rangle \\ &\quad -\frac{B^2}{4} \epsilon_{ij}\epsilon_{kl} \langle \partial_iv^0|(\partial_j\partial_k\hat{H})|\partial_lv^0\rangle \\ &\quad -\frac{B^2}{4} \epsilon_{ij}\epsilon_{kl} \langle \partial_iv^0|(\partial_k\hat{H})|\partial_j\partial_lv^0\rangle \\ &\quad -B\epsilon_{ij} \operatorname{Im} \langle v^1|(\partial_i\hat{H})|\partial_jv^0\rangle + \langle v^1|(\hat{H} - \varepsilon_0)|v^1\rangle. \end{aligned} \quad (10)$$

The last two terms need the first-order state, which is determined in two independent pieces. The normalization $\langle v|\star|v\rangle = 1$ at $O(B)$ fixes the component of $|v^1\rangle$ along $|v^0\rangle$,

$$2 \operatorname{Re} \langle v^0|v^1\rangle + \frac{iB}{2} \epsilon_{ij} \langle \partial_iv^0|\partial_jv^0\rangle = 0 \Rightarrow \operatorname{Re} \langle v^0|v^1\rangle = \frac{B\Omega}{4}, \quad (11)$$

where $\epsilon_{ij}\langle \partial_iv^0|\partial_jv^0\rangle = i\Omega$ is the Berry curvature in our conventions. The imaginary part is left free. The interband part comes from the eigenvalue equation (2) at $O(B)$,

$$(\hat{H} - \varepsilon_0)|v^1\rangle = B\mathcal{M}|v^0\rangle - \frac{iB}{2} \epsilon_{ij} (\partial_i\hat{H})|\partial_jv^0\rangle, \quad (12)$$

which, projected onto a remote band $\langle b^0|$, yields the coefficient of $|b^0\rangle$. In the band basis $\{|b^0\rangle\}$,

$$|v^1\rangle = B\left(\frac{\Omega}{4} + i\alpha\right)|v^0\rangle - \frac{iB}{2} \epsilon_{ij} \sum_{b \neq v} \frac{\langle b^0|\partial_i\hat{H}|\partial_jv^0\rangle}{\varepsilon_b - \varepsilon_0} |b^0\rangle, \quad (13)$$

with α the undetermined real phase left by the normalization.

Projecting Eq. (12) onto $\langle v^1 |$ and taking the real part, with Eq. (11) supplying $\text{Re}\langle v^1 | v^0 \rangle$, converts the Im term of Eq. (10) into

$$-B \epsilon_{ij} \text{Im}\langle v^1 | (\partial_i \hat{H}) | \partial_j v^0 \rangle = \frac{B^2 \mathcal{M} \Omega}{2} - 2 \langle v^1 | (\hat{H} - \varepsilon_0) | v^1 \rangle, \quad (14)$$

while the $\langle v^1 | (\hat{H} - \varepsilon_0) | v^1 \rangle$ term follows from Eq. (13) directly. Only the interband part survives the factor $\hat{H} - \varepsilon_0$, and α drops out,

$$\begin{aligned} \langle v^1 | (\hat{H} - \varepsilon_0) | v^1 \rangle &= \frac{B^2}{4} \epsilon_{ij} \epsilon_{kl} \sum_{b \neq v} \frac{\langle \partial_l v^0 | (\partial_k \hat{H}) | b^0 \rangle \langle b^0 | (\partial_i \hat{H}) | \partial_j v^0 \rangle}{\varepsilon_b - \varepsilon_0}. \end{aligned} \quad (15)$$

Both terms are now free of α . This phase, which is the imaginary part of $\langle v^0 | v^1 \rangle$ up to the factor B , never enters the flat-band dispersion.

One can further simplify the first term of Eq. (10). Differentiating Eq. (5) once more gives

$$\begin{aligned} (\partial_k \partial_i \hat{H}) | v^0 \rangle &= -(\partial_k \hat{H}) | \partial_i v^0 \rangle - (\partial_i \hat{H}) | \partial_k v^0 \rangle \\ &\quad - (\hat{H} - \varepsilon_0) | \partial_k \partial_i v^0 \rangle, \end{aligned} \quad (16)$$

which converts it into matrix elements with single derivatives. Under contraction with the Levi-Civita symbols $\epsilon_{ij} \epsilon_{kl}$, one of the resulting pieces cancels the third term of Eq. (10) exactly. Collecting Eqs. (10)–(16), $\varepsilon^{(2)} = \frac{1}{2} B^2 \mathcal{X}_{\text{raw}}$ with the closed form

$$\begin{aligned} \mathcal{X}_{\text{raw}} &= \mathcal{M} \Omega - \frac{1}{2} \epsilon_{ij} \epsilon_{kl} \langle \partial_l v^0 | (\partial_j \partial_k \hat{H}) | \partial_i v^0 \rangle \\ &\quad + \frac{1}{4} \epsilon_{ij} \epsilon_{kl} \langle \partial_k \partial_i v^0 | (\hat{H} - \varepsilon_0) | \partial_j \partial_l v^0 \rangle \\ &\quad - \frac{1}{2} \epsilon_{ij} \epsilon_{kl} \sum_{b \neq v} \frac{\langle \partial_l v^0 | (\partial_k \hat{H}) | b^0 \rangle \langle b^0 | (\partial_i \hat{H}) | \partial_j v^0 \rangle}{\varepsilon_b - \varepsilon_0}. \end{aligned} \quad (17)$$

Every object on the right-hand side is built from the zero-field wavefunctions. The first term couples the moment to the Berry curvature. The middle terms carry second derivatives of the Hamiltonian and of the band projector. The last is a cross-gap sum over the remote bands.

Unlike the moment, the second-order coefficient is not gauge invariant by itself. This is the order at which the Berry connection $\mathcal{A}_i = i \langle v^0 | \partial_i v^0 \rangle$ first enters. The dependence follows from a covariance of the star-eigenvalue problem. A change of phase convention of the zero-field state, $|v^0\rangle \rightarrow e^{i\phi(\mathbf{\pi})} |v^0\rangle$, is realized on the full solution of Eq. (2) as a star-multiplication from the right, $|v\rangle \rightarrow |v\rangle \star e_\star^{i\phi}$, where $e_\star^{i\phi} = 1 + i\phi + \frac{1}{2}(i\phi) \star (i\phi) + \dots$ is the star-exponential, with inverse $e_\star^{-i\phi}$. The transformed state solves Eq. (2) with the conjugated eigenvalue $\varepsilon \rightarrow e_\star^{-i\phi} \star \varepsilon \star e_\star^{i\phi}$ [24], which to the retained order

is a translation of the symbol,

$$\varepsilon(\mathbf{\pi}) \rightarrow \varepsilon(\mathbf{\pi}_i - B \epsilon_{ij} \partial_j \phi) + O(B^3), \quad (18)$$

the correction being $O(B^3)$ rather than $O(B^2)$ because ε_0 is constant. Applied to $\varepsilon = \varepsilon_0 + B \mathcal{M}(\mathbf{\pi}) + \frac{1}{2} B^2 \mathcal{X}_{\text{raw}}(\mathbf{\pi})$, the translation acts only through the first-order term,

$$\begin{aligned} \varepsilon(\mathbf{\pi}) &\rightarrow \varepsilon_0 + B \mathcal{M}(\mathbf{\pi}) - B^2 \epsilon_{ij} (\partial_i \mathcal{M}) \partial_j \phi \\ &\quad + \frac{1}{2} B^2 \mathcal{X}_{\text{raw}}(\mathbf{\pi}) + O(B^3). \end{aligned} \quad (19)$$

The moment is gauge invariant and the connection transforms as $\mathcal{A}_j \rightarrow \mathcal{A}_j - \partial_j \phi$, so the single combination $B^2 \epsilon_{ij} (\partial_i \mathcal{M}) \mathcal{A}_j$ reproduces the entire variation in Eq. (19). Splitting it off from the second-order term,

$$\mathcal{X}(\mathbf{\pi}) \equiv \mathcal{X}_{\text{raw}}(\mathbf{\pi}) - 2 \epsilon_{ij} (\partial_i \mathcal{M}) \mathcal{A}_j, \quad (20)$$

leaves a remainder that must be unchanged by the gauge transformation: $\mathcal{X}$ is gauge invariant.¹

The split has a compact form, since the connection term is the first-order Taylor term of $B \mathcal{M}$ at a shifted argument. Defining the shifted momentum of Ref. [24],

$$k_i = \pi_i + B \epsilon_{ij} \mathcal{A}_j, \quad (21)$$

absorbs it, and with $\mathcal{X}(\mathbf{k}) = \mathcal{X}(\mathbf{\pi}) + O(B)$ the dispersion reads

$$\mathcal{M}(\mathbf{\pi}) B + \frac{1}{2} \mathcal{X}_{\text{raw}}(\mathbf{\pi}) B^2 = \mathcal{M}(\mathbf{k}) B + \frac{1}{2} \mathcal{X}(\mathbf{k}) B^2 + O(B^3). \quad (22)$$

This is also what makes the invariance of observables manifest. Every physical quantity is a phase-space integral, and Eq. (21) leaves behind only its Jacobian, $d^2 \pi = (1 + B \Omega) d^2 k$ [Eq. (31)]. Integrand and measure are then separately gauge invariant, and the gauge dependence of the integration variable, an area-preserving shift with $\epsilon_{ij} \partial_i \partial_j \phi = 0$, drops out.

The closed form of $\mathcal{X}$ can be shown to be Eq. (17) with every bare derivative of the band state replaced by its covariant counterpart $D_i = \partial_i + i \mathcal{A}_i$:

$$\begin{aligned} \mathcal{X} &= \mathcal{M} \Omega - \frac{1}{2} \epsilon_{ij} \epsilon_{kl} \langle D_i v^0 | (\partial_j \partial_k \hat{H}) | D_l v^0 \rangle \\ &\quad + \frac{1}{4} \epsilon_{ij} \epsilon_{kl} \langle D_k D_i v^0 | (\hat{H} - \varepsilon_0) | D_j D_l v^0 \rangle \\ &\quad - \frac{1}{2} \epsilon_{ij} \epsilon_{kl} \sum_{b \neq v} \frac{\langle D_l v^0 | (\partial_k \hat{H}) | b^0 \rangle \langle b^0 | (\partial_i \hat{H}) | D_j v^0 \rangle}{\varepsilon_b - \varepsilon_0}, \end{aligned} \quad (23)$$

¹ The gauge invariance uses $\partial \varepsilon_0 = 0$. For a dispersive band the same substitution removes the connection from the first-order term, $\varepsilon_0(\mathbf{\pi}) + B \epsilon_{ij} (\partial_i \varepsilon_0) \mathcal{A}_j = \varepsilon_0(\mathbf{k}) + O(B^2)$ [24], but the second-order term then contains $-\epsilon_{ij} (\partial_i \varepsilon_0) \partial_j \alpha$, with α the phase of Eq. (13), which no function of the zero-field connection can absorb. Here α drops out, Eq. (15).

which is gauge invariant term by term. In the numerical checks of Sec. V, $\mathcal{X}$ is evaluated pointwise in the parallel-transport gauge, $\mathcal{A} = 0$ at the evaluation point, in which Eq. (23) reduces to Eq. (17), with the derivatives of $|v^0\rangle$ and $\hat{H}$ taken by finite differences.

The Jacobian of the substitution (21), $1 + B\Omega$, is a direct consequence of gauge invariance under the Moyal product. The same phase-space volume factor is known from the semiclassical approximation [17, 29, 30], and here it applies at $O(B^2)$ and for an exactly flat band. The assembled dispersion is then gauge invariant term by term:

$$\varepsilon(\mathbf{k}; B) = \varepsilon_0 + \mathcal{M}(\mathbf{k}) B + \frac{1}{2} \mathcal{X}(\mathbf{k}) B^2 + O(B^3). \quad (24)$$

The second-order dispersion of Ref. [31], obtained from the same Moyal expansion, reduces on an exactly flat band and in the present notation to

$$\mathcal{X}_{\text{PHSB}} = \mathcal{X} - \mathcal{M}\Omega + \frac{1}{2} \varepsilon_0 \Omega^2. \quad (25)$$

The last term violates gauge invariance under time-dependent gauge transformations. A gauge function linear in time leaves the vector potential unchanged and shifts the scalar potential, and with it $\hat{H}$ and every band energy, by a constant c . Under this shift of the energy zero, $\hat{H} \rightarrow \hat{H} + c$, the last term changes by $\frac{1}{2} c \Omega^2$, whose zone integral is nonzero for any band with nonvanishing Berry curvature, although no energy difference in the problem is altered. Every term of Eq. (23) is unchanged, since the Hamiltonian enters only through $\hat{H} - \varepsilon_0$ and its derivatives.

D. The expansion parameter

For a dispersive band the ratio $\hbar\omega_c/E_F$ of the cyclotron energy to the Fermi energy is an obvious small parameter in the semiclassical expansion. A flat band has neither scale; the effective mass diverges, so the cyclotron energy $\hbar\omega_c \propto B/m^*$ vanishes at leading order, and the Fermi energy measured from the band is zero. The expansion parameter is instead set by the Moyal product.

Every order of the star product (2) inserts one factor of B and one derivative pair $\epsilon_{ij}\partial_i\partial_j$, so the expansion runs formally in $B(\partial_k)^2$, the field times the square of a length, whose value is set by the momentum dependence on which the derivatives act. On a flat band ε_0 has none, so every derivative lands on the wavefunctions, and their momentum dependence is measured by the quantum metric $g_{ij} = \text{Re} \langle \partial_i v^0 | Q | \partial_j v^0 \rangle$, with $Q = 1 - |v^0\rangle\langle v^0|$. We define the geometric length ξ of the band by $\xi^2 = 4 \max_{\mathbf{k}} \lambda_{\max}(g)$, where $\lambda_{\max}(g)$ is the larger eigenvalue of g . The factor of four calibrates ξ to the length $v/|m|$ of a massive Dirac cone with velocity v and half-gap $|m|$, evaluated at the end of this subsection. Every projected derivative obeys $\|Q|\partial_i v^0\rangle\| \leq \xi$.

Each wavefunction derivative in the expansion therefore contributes a factor of at most ξ , and the dimensionless expansion parameter is

$$x = B\xi^2 = \left(\frac{\xi}{\ell_B}\right)^2 \ll 1, \quad (26)$$

the flux through the geometric area ξ^2 , equivalently the squared ratio of ξ to the magnetic length $\ell_B = B^{-1/2}$.

The length ξ has a direct physical reading. For a band that admits exponentially localized Wannier functions, the zone average of $\text{tr} g$ is the gauge-invariant part of the quadratic spread of the maximally localized Wannier function [32]. Thus ξ measures the linear size of the most localized orbital the band supports, and x is the flux through that orbital.

The closed forms verify this counting. Equation (7) carries two projected derivatives (the $(\hat{H} - \varepsilon_0)$ insertion retains only $Q|\partial_i v^0\rangle$, each bounded by ξ) and one gap-scale insertion, and every term of Eq. (17) carries two more wavefunction derivatives, each contributing a further factor of ξ . Hence

$$|\mathcal{M}| \sim \Delta \xi^2, \quad |\mathcal{X}| \sim \Delta \xi^4 \implies \frac{\frac{1}{2} \mathcal{X} B^2}{\mathcal{M} B} \sim B \xi^2, \quad (27)$$

and the n -th Moyal order is of size $\Delta(B\xi^2)^n$.² Near the gap minimum of a massive Dirac cone with velocity v and half-gap $|m|$, the length defined above is $\xi = v/|m|$, and the scaling (27) is realized exactly,

$$\mathcal{M} = -\frac{1}{2} |m| \xi^2, \quad (28)$$

with the half-gap in place of Δ .

III. THE ORBITAL MAGNETIC SUSCEPTIBILITY OF A FLAT BAND

At the noninteracting level, the susceptibility of a flat band is contained in the gauge-invariant multiband formulas [18, 20] as the limit of vanishing dispersion. This section derives it along a different route, from the field-generated dispersion of Sec. II, and proves that the two routes agree. Interacting extensions exist. Ref. [22] formulates the interacting problem in the noncommutative kinetic momentum, the algebra of Sec. II, and obtains the

² This estimate uses a single energy scale Δ . The gap appears directly only in $\mathcal{M}$ and in the energy denominator of the cross-gap term of Eq. (17), dominated by the nearest remote band. That term and the band-curvature term $\partial_j \partial_k \hat{H}$ also involve the velocities of the remote sector, $\langle b^0 | \partial_i \hat{H} | c^0 \rangle$, which the flat-band metric does not constrain, and we take them to be of the same scale, $\sim \Delta \xi$. If the nearest remote band is anomalously dispersive, its velocity scale replaces $\Delta \xi$ there and enlarges the effective expansion parameter accordingly.

orbital magnetization at first order in B . Ref. [21], working from the grand potential at finite field, carries the interacting susceptibility to second order at self-consistent mean-field level, as a band functional supplemented by the feedback of the field-dependent mean field. Ref. [23] shows that interactions render the Landau diamagnetic susceptibility of a two-dimensional Fermi liquid nonanalytic in temperature and in-plane field.

The grand potential per unit area is the phase-space trace of $\mathcal{F}_T(\hat{H})$, the trace per unit area being $\int d^2\pi/(2\pi)^2$ of the symbol [28], with $\mathcal{F}_T(\varepsilon) = -T \ln(1 + e^{-(\varepsilon-\mu)/T})$. Two steps convert the trace to an integral. First, the trace of a star product of two symbols is the plain integral of their ordinary product,

$$\int \frac{d^2\pi}{(2\pi)^2} a \star b = \int \frac{d^2\pi}{(2\pi)^2} a b, \quad (29)$$

because every insertion of the star exponential (2) beyond the zeroth is a total derivative under the integral. Second, the symbol of $\mathcal{F}_T(\hat{H})$ is $\mathcal{F}_T(\varepsilon)$ up to corrections from the star products among the factors of ε in the power series of $\mathcal{F}_T$. In the star product (2) every power of B comes with one derivative acting on each of the two factors, so each correction differentiates at least two factors of ε . A flat band has no zero-field dispersion, so a differentiated factor of ε is itself $O(B)$. The lowest correction, one explicit power of B times two such factors, is therefore $O(B^3)$, beyond the order retained. With the dispersion of Sec. II C,

$$\Xi(B) = \int \frac{d^2\pi}{(2\pi)^2} \mathcal{F}_T(\varepsilon_0 + \mathcal{M}(\pi)B + \frac{1}{2}\mathcal{X}_{\text{raw}}(\pi)B^2) + O(B^3), \quad (30)$$

with flat measure but the gauge-dependent coefficient. The identity (22) replaces $\mathcal{X}_{\text{raw}}(\pi)$ by $\mathcal{X}(\mathbf{k})$ in the argument, and its Jacobian, the Berry-measure factor $1+B\Omega$, affects the measure,

$$\Xi(B) = \int \frac{d^2k}{(2\pi)^2} (1+B\Omega) \mathcal{F}_T(\varepsilon(\mathbf{k}; B)). \quad (31)$$

Every object in Eq. (31) is gauge invariant. Differentiating at $B=0$ gives the magnetization and susceptibility,

$$M = -\partial_B \Xi|_0 = -\int \frac{d^2k}{(2\pi)^2} [\mathcal{M}f + \Omega \mathcal{F}_T]_{\varepsilon_0}, \quad (32)$$

$$\begin{aligned} \chi_{\text{thermo}} &= -\partial_B^2 \Xi|_0 \\ &= -\int \frac{d^2k}{(2\pi)^2} [(\mathcal{X} + 2\mathcal{M}\Omega)f + \mathcal{M}^2 f']_{\varepsilon_0}, \end{aligned} \quad (33)$$

with $f = \mathcal{F}_T'$ the Fermi function and all integrands evaluated on the flat band. The magnetization is the thermally weighted moment plus the Berry-measure correction, and the susceptibility collects $\mathcal{X}$, the moment-curvature cross term, and the Curie-like $\mathcal{M}^2 f'$ term of the field-generated moments.

Equation (33) agrees with the gauge-invariant susceptibility of Ref. [18] applied to a flat band: the exact trace formula

$$\begin{aligned} \chi_{\text{thermo}} &= -\frac{1}{12} \frac{\text{Im}}{\pi} \int dE f(E) \int \frac{d^2k}{(2\pi)^2} \text{Tr} \{ Gh^{xx} Gh^{yy} \\ &\quad - Gh^{xy} Gh^{xy} - 4(Gh^x Gh^x Gh^y Gh^y \\ &\quad - Gh^x Gh^y Gh^x Gh^y) \}, \end{aligned} \quad (34)$$

with $G = (E - h + i0)^{-1}$, $h^a \equiv \partial h / \partial k_a$, and $h^{ab} \equiv \partial^2 h / \partial k_a \partial k_b$, where $h(\mathbf{k})$ is the zero-field Bloch Hamiltonian of Sec. II A, here a function of the crystal momentum, and the indices a, b run over x, y . In Appendix A we reduce Eq. (34) at an arbitrary isolated, nondegenerate flat band and recover the structure and the coefficients of Eq. (33): the same two thermal weights $f(\varepsilon_0)$ and $f'(\varepsilon_0)$ and no others, a Curie coefficient that matches $-\mathcal{M}^2$ pointwise for any number of bands, and an f -coefficient that matches $-\mathcal{X} + 2\mathcal{M}\Omega$ after zone integration. The susceptibility of an isolated flat band is thus fixed by three functions of the zero-field wavefunctions ($\mathcal{M}$, $\mathcal{X}$, Ω), in agreement with the established gauge-invariant response theory [18, 20, 21].

The procedure of this section is not restricted to the flat band. The same two steps, expand the dispersion in the kinetic momentum and differentiate the grand potential, apply to a dispersive band, where $\varepsilon_0(\pi)$ is momentum dependent in the argument of $\mathcal{F}_T$ [33]. They apply also to interacting systems treated at self-consistent mean-field level, where the expansion runs over the Hartree-Fock bands and the field dependence of the mean field enters as a feedback term [21, 22].

For a two-band model the sums over remote bands can be carried out in closed form, and we record the result here for later use in Sec. V. Evaluating the cross-gap sum of Eq. (17), which for the two-band problem runs over a single remote band, gives

$$\mathcal{X}(\mathbf{k}) = \frac{1}{4} U_2(\mathbf{k}) - \Omega^2(\mathbf{k}), \quad U_2 \equiv \hat{\mathbf{d}}_{xx} \cdot \hat{\mathbf{d}}_{yy} - \hat{\mathbf{d}}_{xy} \cdot \hat{\mathbf{d}}_{xy}, \quad (35)$$

where $\hat{\mathbf{d}}$ is the unit Bloch vector of the model, $\hat{H} = \hat{\mathbf{d}} \cdot \boldsymbol{\sigma}$, and subscripts denote momentum derivatives. U_2 is a total divergence, so its zone integral vanishes. The full susceptibility then follows from the trace-formula reduction of Appendix A. The remote-band (frozen) contribution vanishes, because the resolvent has no bands beyond the pair whose pole terms are kept, and assembling the two bands ($\varepsilon \rightarrow -\varepsilon$ sends $\Omega \rightarrow -\Omega$ and $\mathcal{X} \rightarrow -\mathcal{X}$ with $\mathcal{M}$ unchanged) gives

$$\begin{aligned} \chi_{\text{thermo}}(\mu, T) &= -I [f(-1) - f(+1) + f'(-1) + f'(1)], \\ I &\equiv \int \frac{d^2k}{(2\pi)^2} \Omega^2, \end{aligned} \quad (36)$$

a single wavefunction integral carrying the full (μ, T) -dependence in the bracket of Fermi factors.

IV. THE DE HAAS–VAN ALPHEN EFFECT OF A FLAT BAND

The response of Sec. III is smooth in B and therefore contains no de Haas–van Alphen oscillations. The reason lies in a single step. Converting the trace of $\mathcal{F}_T(\hat{H})$ into the kinetic-momentum integral (30) replaced a sum over the discrete spectrum of the star-eigenvalue problem by the integral of a smooth symbol, order by order in B . The discreteness of the spectrum survives in that conversion only through corrections nonanalytic in B , which vanish at every order of the expansion. One route to the oscillations is therefore to redo the conversion carefully, keeping the level structure of the star spectrum and restoring the nonanalytic terms by Poisson summation [3, 4, 15, 34]. There is an alternative. At fixed chemical potential the oscillations are carried entirely by the low-energy excitations near μ , and for a Fermi system those excitations admit an exact bosonized description. We take the second route.

For the flat band the low-energy structure is generated by the field itself. Pin $\mu = \varepsilon_0$ and let $B > 0$. States with $\mathcal{M} < 0$ are filled, states with $\mathcal{M} > 0$ are empty, and the Fermi surface is the zero contour $\mathcal{C}_0$ of Eq. (8). We assume $\mathcal{M}$ changes sign and $\mathcal{C}_0$ exists. Ref. [12] showed that the Landau levels of an isolated flat band, with spacing $\sim B^2$, are those of the effective dispersion $B\mathcal{M}(\mathbf{k})$, the first-order dispersion of Sec. IIB. One thus expects quantum oscillations.

It is not guaranteed that the semiclassical rules of a dispersive band apply to an emergent Fermi surface. In a metal the quantization phase is the Maslov one-half plus the Berry phase of a zero-field orbit. Here the orbit itself exists only at finite field, its dispersion carries the second-order coefficient $\mathcal{X}$ alongside the moment, and how the phase is related to the Berry curvature and to these other terms of the field-generated dispersion is not settled by analogy. The derivation below decides it without semiclassical input. No quantization rule is assumed, and the phase comes out of the topology of the bosonized theory. It holds for an isolated, nondegenerate flat band whose moment $\mathcal{M}(\mathbf{k})$ is smooth with a single closed $\mathcal{M} = 0$ contour.

A. Bosonized action and projection to the flat band

Our derivation uses coadjoint-orbit bosonization, proposed in Ref. [25] as an exact bosonic description of a Fermi surface in terms of its shape fluctuations. It was applied to the de Haas–van Alphen effect of a Fermi gas and a Fermi liquid in Ref. [26] and extended to multiband systems in Ref. [24]. The multiband version is the method we apply to the flat band. For a single Fermi contour the fluctuations organize into one chiral boson $\phi(\theta, t)$, with θ a coordinate along the contour, defined in Sec. IV B. Because the boson is compact, its configuration space splits into topological sectors labelled by an

integer winding number, which physically represents the number of particles added to the contour.

The action of the multiband construction, obtained in Ref. [24] from the fermion path integral over particle–hole coherent states, is

$$S = \int dt \frac{d\mathbf{x} d\mathbf{p}}{(2\pi)^2} \text{Tr} \left[\hat{f}_0 \star \hat{U}^{-1} \star i\partial_t \hat{U} - \hat{f} \star \hat{H} \right], \quad (37)$$

where every hatted object is a matrix in the band (orbital) space and Tr traces over that space. The reference distribution $\hat{f}_0$ is the Wigner transform of the ground-state one-particle density matrix $\rho_0(\mathbf{x}, \mathbf{x}')$,

$$\hat{f}_0(\mathbf{x}, \mathbf{p}) = \int d^2s e^{-i\mathbf{p}\cdot\mathbf{s}} \rho_0\left(\mathbf{x} + \frac{\mathbf{s}}{2}, \mathbf{x} - \frac{\mathbf{s}}{2}\right), \quad (38)$$

a matrix in the orbital indices. The unitary $\hat{U} = e^{i\hat{\phi}}$, with $\hat{\phi}(\mathbf{x}, \mathbf{p})$ a Hermitian matrix symbol, generates the particle–hole rotations of the ground state, and $\hat{f} = \hat{U} \star \hat{f}_0 \star \hat{U}^{-1}$ is the rotated distribution. The $\star$ in Eq. (37) is the phase-space Moyal product

$$\star = \exp \left[\frac{i}{2} (\overleftarrow{\partial}_{\mathbf{x}} \cdot \overrightarrow{\partial}_{\mathbf{p}} - \overleftarrow{\partial}_{\mathbf{p}} \cdot \overrightarrow{\partial}_{\mathbf{x}}) \right], \quad (39)$$

defined by gradients in $\mathbf{x}$ and $\mathbf{p}$. It is not the momentum-space product of Eq. (2), which acts on functions of $\boldsymbol{\pi}$ alone. No expansion has been made. The action (37) reproduces the full fermion partition function restricted to the particle–hole coherent states $\hat{U}|G\rangle$, with $|G\rangle$ the ground state [24].

Two changes of variables bring the action to the flat band [24, 35]. The first is $(\mathbf{x}, \mathbf{p}) \rightarrow (\mathbf{R}, \boldsymbol{\pi})$, with $\boldsymbol{\pi} = \mathbf{p} - \mathbf{A}(\mathbf{x})$ the kinetic momentum ($\mathbf{A}$ is the electromagnetic vector potential) and $R_i = x_i + \epsilon_{ij}\pi_j/B$ the guiding center. The two sets $\star$ -commute, and the product (39) factorizes into a $\boldsymbol{\pi}$ -sector product, which is Eq. (2), and an $\mathbf{R}$ -sector product of the same form with $B \rightarrow -1/B$, where for $[f, g]_\star \equiv f \star g - g \star f$,

$$[\pi_i, \pi_j]_\star = iB\epsilon_{ij}, \quad [R_i, R_j]_\star = -\frac{i}{B}\epsilon_{ij}, \quad [R_i, \pi_j]_\star = 0 \quad (40)$$

[35]. We project onto the flat band with the same $\star$ -eigenstates that built Eq. (24). Every matrix symbol is expanded in the star-eigenbasis of the bands, and the flat-band diagonal element is kept,

$$f_0(\mathbf{k}) = \langle v | \star \hat{f}_0 \star | v \rangle, \quad \phi(\mathbf{R}, \mathbf{k}) = \langle v | \star \hat{\phi} \star | v \rangle. \quad (41)$$

The off-band elements of $\hat{\phi}$ create particle–hole pairs across the gap. Their contribution to the action carries the energy denominator Δ , while every retained scale is a power of the field, so they enter beyond the retained order and the flat-band scalars (41) are the fluid variables. The second change is $(\mathbf{R}, \boldsymbol{\pi}) \rightarrow (\mathbf{R}, \mathbf{k})$, by the shifted momentum of Eq. (21) [24]. In the conventions of Sec. IIC the connection obeys $\epsilon_{ij}\partial_i A_j = -\Omega$, so the Jacobian of Eq. (21) is $\det(\partial k / \partial \pi) = 1 + B\epsilon_{ij}\partial_i A_j = 1 - B\Omega$,

and the measure transforms as $d^2\pi = (1 + B\Omega) d^2k$ to the retained order, the measure of Eq. (31). The momentum-sector bracket carries the inverse factor, using $[\pi_i, f]_\star = iB \epsilon_{il} \partial_l f$,

$$\begin{aligned} [k_i, k_j]_\star &= [\pi_i, \pi_j]_\star + B \epsilon_{jm} [\pi_i, \mathcal{A}_m]_\star + B \epsilon_{il} [\mathcal{A}_l, \pi_j]_\star \\ &= iB \epsilon_{ij} (1 + B \epsilon_{lm} \partial_l \mathcal{A}_m) = \frac{iB \epsilon_{ij}}{1 + B\Omega} \end{aligned} \quad (42)$$

to the retained order. The result is a single-band fluid [24]:

$$S = \int_{t, \mathbf{R}, \mathbf{k}} (1 + B\Omega) \left[f_0(\mathbf{k}) U^{-1} \star i \partial_t U - f \star \varepsilon(\mathbf{k}; B) \right], \quad (43)$$

where $f_0 = \Theta(\mu - \varepsilon)$, with Θ the Heaviside step function, is the zero-temperature value of the projected occupation (41), and $\star$ is the product in the $(\mathbf{R}, \mathbf{k})$ variables. One ingredient extends Ref. [24]. The dispersion entering Eq. (43) is the full second-order symbol of Eq. (24). The analysis of Ref. [24] stops at first order. For the flat band the second-order term survives in the phase.

B. Chiral boson on the emergent contour

The reduction to the contour proceeds in two steps. Magnetic translation symmetry reduces the two-dimensional boson to N_Φ copies, and the expansion around the ground state localizes each copy on the emergent Fermi contour.

By magnetic translation symmetry it suffices to keep configurations invariant under magnetic translations [24], the symmetry-breaking configurations contributing only Gaussian factors smooth in B , and the invariant ones carry no dependence on the guiding center. The guiding-center plane is not a passive label. By the algebra (40) it is a noncommutative plane carrying $N_\Phi = BL^2/2\pi$ states, with L^2 the system area, so the $\mathbf{R}$ -integral in Eq. (43) is a trace over those states, and an $\mathbf{R}$ -independent configuration contributes one identical copy per state. Thus $\phi(\mathbf{R}, \mathbf{k})$ reduces to N_Φ copies $\phi_i(\mathbf{k})$, one per magnetic state. Recall that ϕ enters the action through $U = e^{i\phi}$, so an expansion in ϕ_i is an expansion of Eq. (43) around the ground state. To quadratic order the two factors of Eq. (43) expand as

$$\begin{aligned} U^{-1} \star i \partial_t U &= -\dot{\phi}_i + \frac{i}{2} [\phi_i, \dot{\phi}_i]_\star + O(\phi_i^3), \\ f &= f_0 + i [\phi_i, f_0]_\star - \frac{1}{2} [\phi_i, [\phi_i, f_0]_\star]_\star + O(\phi_i^3), \end{aligned} \quad (44)$$

where, by the bracket (42), to the retained order $[a, b]_\star = \frac{iB}{1+B\Omega} \epsilon_{lm} \partial_l a \partial_m b$, with l, m running over k_x, k_y . For non-winding configurations the terms linear in ϕ_i integrate to zero. The ground state extremizes the action. The winding sector is kept below. Two structures organize the

quadratic terms. First, every single commutator with f_0 is supported on the emergent contour, where f_0 jumps:

$$[\phi_i, f_0]_\star = -\frac{iB}{1+B\Omega} \delta(\varepsilon(\mathbf{k}; B) - \mu) \epsilon_{lm} \partial_l \phi_i \partial_m \varepsilon, \quad (45)$$

and $\epsilon_{lm} \partial_l \phi_i \partial_m \varepsilon$ is $|\partial_{\mathbf{k}} \varepsilon|/|\partial_\theta \mathbf{k}|$ times the derivative of ϕ_i along the contour, where θ parametrizes the contour counterclockwise, $\mathbf{k}(\theta)$. Terms carrying this support involve only the contour modes $\phi_i(\theta)$, the gapless directions. Second, the kinetic quadratic term $f_0 \frac{i}{2} [\phi_i, \dot{\phi}_i]_\star$ is smooth in f_0 , and one integration by parts splits it into a contour part, carrying the jump of f_0 , and a residual curl density $\frac{1}{2} \epsilon_{lm} \partial_l \partial_m \phi_i \dot{\phi}_i$, which vanishes for smooth ϕ_i and survives only on the singular winding configurations treated in Sec. IV C. Assembling the pieces, and reducing the energy term by cyclicity of the star product under the integral, $\int [\phi_i, [\phi_i, f_0]_\star]_\star \varepsilon = -\int [\phi_i, f_0]_\star [\phi_i, \varepsilon]_\star$, the action splits into two pieces [24], $S = \sum_{i=1}^{N_\Phi} (S_{\text{cb}} + S_{\text{w}})$, with

$$S_{\text{cb}} = \int \frac{dt d\theta}{4\pi} \partial_\theta \phi_i \left(\dot{\phi}_i - \omega(\theta) \partial_\theta \phi_i \right), \quad (46)$$

$$S_{\text{w}} = \int \frac{dt d^2k}{2\pi} f_0(\mathbf{k}) \left[-\frac{1+B\Omega}{B} \dot{\phi}_i + \frac{1}{2} \epsilon_{lm} \partial_l \partial_m \phi_i \dot{\phi}_i \right]. \quad (47)$$

The first term of Eq. (46) is the contour part of the kinetic quadratic term. The second collects the reduced energy term, both single commutators evaluated by Eq. (45), with $\omega(\theta)$ the local drift frequency of the field-generated band along the contour,

$$\omega(\theta) = \frac{B |\partial_{\mathbf{k}} \varepsilon|}{(1+B\Omega) |\partial_\theta \mathbf{k}|} \Big|_{\varepsilon=\mu}. \quad (48)$$

Equation (46) is a chiral boson propagating along the contour [36]. Equation (47) collects the smooth f_0 -weighted terms: the zeroth-order time derivative and the curl density, written with explicit indices. The term (47), integrated over the occupied region, does not affect the equations of motion. It is nonzero only on winding configurations, and it is what carries the oscillation physics.

The boson is compact. Its mode expansion on the contour,

$$\phi_i(\theta) = q_i + w_i \theta + \sum_{n \neq 0} \frac{a_{i,n}}{\sqrt{|n|}} e^{in\theta}, \quad q_i \simeq q_i + 2\pi, \quad (49)$$

contains a compact zero mode q_i and an integer winding number w_i . The winding is the physical charge added to the contour at magnetic state i . Expanding the conjugation to first order in ϕ_i and applying Eq. (45), the change in occupation is

$$\begin{aligned} \delta n_i &= \int \frac{d^2k}{2\pi B} (1 + B\Omega) (e^{i\phi_i} \star f_0 \star e^{-i\phi_i} - f_0) \\ &= \int \frac{d^2k}{2\pi B} (1 + B\Omega) i [\phi_i, f_0]_\star + O(\phi_i^2) \\ &= -\int_{-\pi}^{\pi} \frac{d\theta}{2\pi} \partial_\theta \phi_i = -w_i, \end{aligned} \quad (50)$$

where θ increases counterclockwise, the orientation used throughout, and the higher orders in ϕ_i integrate to zero on a winding configuration. The signs are written for an electron-like contour, and on a hole-like contour both this sign and that of the winding term flip, and the two flips cancel in the zero-mode action derived next. Through its winding sector the particle number can change, as it must at fixed μ .

C. The zero-mode sector

Inserting the mode expansion (49) into $S_{\text{cb}} + S_{\text{w}}$ separates the action into a Fock sector of the oscillators $a_{i,n}$ and a zero-mode sector of the pairs (q_i, w_i) . For the Fermi gas and the Fermi liquid, Ref. [26] showed that the de Haas–van Alphen oscillations are carried by the zero-mode sector alone. The same division holds here, and for the same reason. The Fock oscillators $a_{i,n}$ enter the action (46) with no coupling to the occupied-state count $n(\mu, B)$ introduced below, so their Gaussian integral is a prefactor of the partition function, smooth in B . The B -periodic structure enters only through the winding numbers, the discrete charge transfers of Eq. (50), which shift the occupation against the count. We therefore keep the pairs (q_i, w_i) and derive their action.

Evaluated at $\partial_\theta \phi_i = w_i$ and $\dot{\phi}_i = \dot{q}_i$, the first term of Eq. (46) gives $\frac{1}{2} w_i \dot{q}_i$. The second term of Eq. (47) is evaluated on the winding configuration extended into the occupied region, a field whose angular winding is w_i around an interior point; for any such extension

$$\int_{\varepsilon < \mu} \frac{d^2 k}{2\pi} \frac{1}{2} \epsilon_{lm} \partial_l \partial_m \phi_i \dot{q}_i = \frac{1}{2} w_i \dot{q}_i, \quad (51)$$

since $\epsilon_{lm} \partial_l \partial_m \phi_i$ integrates, by Stokes' theorem, to the circulation $\oint \partial_\theta \phi_i d\theta = 2\pi w_i$ of the boundary value, independent of the interior extension. Together the two halves give the kinetic term $w_i \dot{q}_i$. The first term of Eq. (47) contributes $-n(\mu, B) \dot{q}_i$, with

$$n(\mu, B) = \frac{1}{2\pi B} \int_{\varepsilon < \mu} (1 + B\Omega) d^2 k, \quad (52)$$

the occupied phase-space volume per magnetic state in the measure of Eq. (31). The velocity term of Eq. (46) is quadratic in $\partial_\theta \phi_i$. It is also the only term that depends on how the contour is parametrized, since the kinetic term $\partial_\theta \phi_i \dot{\phi}_i$ is reparametrization invariant (the Jacobian of $d\theta$ cancels that of ∂_θ) and the winding numbers are integrals of $\partial_\theta \phi_i d\theta$. Relabeling θ so that ω becomes uniform therefore changes nothing else. The new parameter still spans 2π , so a uniform frequency has circuit time $2\pi/\hbar\tilde{\omega}_{\text{eff}}$, and equating this to the invariant circuit time $\oint d\theta/\omega(\theta)$ gives $\hbar\tilde{\omega}_{\text{eff}} = 2\pi[\oint d\theta/\omega(\theta)]^{-1}$. Along the drift $\dot{\theta} = \omega(\theta)$, so $\oint d\theta/\omega(\theta)$ is the drift period T around the contour and $\hbar\tilde{\omega}_{\text{eff}} = 2\pi\hbar/T$ is the level spacing of an orbit of period T . In the uniform parametriza-

tion the winding mode has $\partial_\theta \phi_i = w_i$, and the velocity term contributes $-\frac{\hbar\tilde{\omega}_{\text{eff}}}{2} w_i^2$. The zero-mode action is therefore

$$S_{\text{zero}} = \sum_{i=1}^{N_\Phi} \int dt \left[w_i \dot{q}_i - n(\mu, B) \dot{q}_i - \frac{\hbar\tilde{\omega}_{\text{eff}}}{2} w_i^2 \right], \quad (53)$$

and its two coefficients, the count $n(\mu, B)$ and the drift frequency $\hbar\tilde{\omega}_{\text{eff}}$, remain to be evaluated.

The count separates into area and Berry flux,

$$n(\mu, B) = \frac{A_{\text{occ}}(B)}{2\pi B} + \frac{1}{2\pi} \int_{\varepsilon < \mu} \Omega d^2 k, \quad (54)$$

where the flux integral runs over the occupied region $\{\mathbf{k} : \varepsilon(\mathbf{k}; B) < \mu\}$ and $A_{\text{occ}}(B)$ is its area, field-dependent through the second-order dispersion. With $\mu = \varepsilon_0$, the Fermi contour sits at $\mathcal{M} = -\frac{B}{2}\mathcal{X}|_{\mathcal{C}_0} + \dots$. Each point of $\mathcal{C}_0$ is displaced along $\nabla\mathcal{M}/|\nabla\mathcal{M}|$ by the distance $-\frac{B}{2}\mathcal{X}/|\nabla\mathcal{M}|$. Each boundary element transfers the strip of width $\frac{B}{2}\mathcal{X}/|\nabla\mathcal{M}|$ out of the occupied region where $\mathcal{X} > 0$ and into it where $\mathcal{X} < 0$, for either contour family, so

$$A_{\text{occ}}(B) = A_{\text{occ}}(0) - \frac{B}{2} \oint_{\mathcal{C}_0} \frac{\mathcal{X}}{|\nabla\mathcal{M}|} d\ell. \quad (55)$$

The zero-field value is fixed by the area $A_{\mathcal{M}}$ enclosed by $\mathcal{C}_0$. One has $A_{\text{occ}}(0) = A_{\mathcal{M}}$ for an electron-like contour, whose enclosed $\mathcal{M} < 0$ side is occupied, and $A_{\text{occ}}(0) = 4\pi^2 - A_{\mathcal{M}}$ for a hole-like one, whose enclosed pocket $P = \{\mathbf{k} : \mathcal{M}(\mathbf{k}) > 0\}$ is unoccupied. In the hole-like case the occupied-region flux is $2\pi C - \int_P \Omega d^2 k + O(B)$, with C the Chern number of the band, and the count becomes

$$n(\mu, B) = \left(\frac{2\pi}{B} + C \right) - \frac{4\pi^2 - A_{\text{occ}}(B)}{2\pi B} - \frac{1}{2\pi} \int_P \Omega d^2 k + O(B), \quad (56)$$

where $2\pi/B$ is the number of flux quanta per Brillouin zone. Every quantitative statement below is made at the fluxes $B = 2\pi/q$ with integer q , where $2\pi/B + C$ is an integer and drops from every harmonic. The area, the Berry flux, and the $\mathcal{X}$ correction enter the oscillations only through the single function $n(\mu, B)$.

The frequency follows by substituting Eq. (48) into the period; with $d\ell = |\partial_\theta \mathbf{k}| d\theta$ as the line element of the contour,

$$\frac{2\pi}{\hbar\tilde{\omega}_{\text{eff}}} = \oint \frac{d\theta}{\omega(\theta)} = \oint_{\varepsilon=\mu} \frac{(1 + B\Omega) d\ell}{B |\partial_{\mathbf{k}} \varepsilon|}. \quad (57)$$

On the contour the gradient is set by the moment alone, $\partial_{\mathbf{k}} \varepsilon = B \nabla \mathcal{M} + O(B^2)$ by Eq. (24), and the measure factor enters at the same order, so to leading order

$$\hbar\tilde{\omega}_{\text{eff}} = \frac{2\pi B^2}{\oint_{\mathcal{C}_0} d\ell/|\nabla\mathcal{M}|}, \quad (58)$$

quadratic in B . One power is explicit in Eq. (48), and $|\partial_{\mathbf{k}}\varepsilon|$ carries the second, the dispersion being itself of first order. The B^2 scale of the spacing was identified in Ref. [12] by applying the Onsager rule to the effective dispersion $B\mathcal{M}(\mathbf{k})$, whose effective mass grows as $1/B$. Equation (58) supplies the coefficient in closed form. The period integral in Eq. (57) is exactly the μ derivative of the count. Raising μ in Eq. (52) advances the contour locally by $d\mu/|\partial_{\mathbf{k}}\varepsilon|$, each element weighted by the measure $1 + B\Omega$, so that $\oint_{\varepsilon=\mu} (1 + B\Omega) d\ell / (B|\partial_{\mathbf{k}}\varepsilon|) = 2\pi dn/d\mu$ and $\hbar\tilde{\omega}_{\text{eff}} = (dn/d\mu)^{-1}$ to all orders in B , the inverse density of states per magnetic state. This is the expected relation between level spacing and density of states.

With both coefficients evaluated, the spectrum follows. For each magnetic state, Eq. (53) is the quantum mechanics of a particle on a ring threaded by a flux [24, 37]. Here q_i is the compact coordinate, the count $n(\mu, B)$ plays the role of the flux, and $\hbar\tilde{\omega}_{\text{eff}}$ plays the role of the charging energy. The coefficient of $\dot{q}_i$ in Eq. (53) identifies the momentum conjugate to q_i as $w_i - n(\mu, B)$, and compactness of q_i , of period 2π , quantizes this momentum to integer eigenvalues l_i . On such an eigenstate $w_i = l_i + n(\mu, B)$ (integer windings label configurations of the path integral, not eigenstates), and the quadratic term of Eq. (53) is

$$\frac{\hbar\tilde{\omega}_{\text{eff}}}{2} w_i^2 = \frac{\hbar\tilde{\omega}_{\text{eff}}}{2} (l_i + n(\mu, B))^2. \quad (59)$$

The partition function per magnetic state is

$$Z_{\text{zero}} = \sum_{l \in \mathbb{Z}} \exp\left[-\frac{\hbar\tilde{\omega}_{\text{eff}}}{2T} (l + n(\mu, B))^2\right], \quad (60)$$

periodic in the count with unit period. The discreteness of the spectrum enters through the integer eigenvalues of the conjugate momentum, without a Landau-level sum.

D. Free energy of the zero mode: LK series and the quantization phase

The oscillatory grand potential is the free energy of the zero-mode sector, $\delta\Xi_{\text{osc}} = -(N_{\Phi}/L^2) T \ln Z_{\text{zero}}$, with $N_{\Phi} = BL^2/2\pi$ the number of magnetic states. The B -periodic structure is extracted by Poisson resummation of the ladder (60),

$$Z_{\text{zero}} = \sqrt{\frac{2\pi T}{\hbar\tilde{\omega}_{\text{eff}}}} \sum_{p \in \mathbb{Z}} e^{-2\pi^2 p^2 T / \hbar\tilde{\omega}_{\text{eff}}} e^{2\pi i p n(\mu, B)}, \quad (61)$$

which trades the winding sum for a sum over harmonics. Define

$$X_p = \frac{2\pi^2 p T}{\hbar\tilde{\omega}_{\text{eff}}}, \quad (62)$$

so the Gaussian weight of harmonic p in Eq. (61) is $e^{-pX_p} = e^{-p^2 X_1}$. The p -sum of Eq. (61) is a Jacobi theta

function,

$$Z_{\text{zero}} = \sqrt{\frac{2\pi T}{\hbar\tilde{\omega}_{\text{eff}}}} \theta_3(\pi n(\mu, B), e^{-X_1}), \quad (63)$$

$$\theta_3(\zeta, e^{-X_1}) = \sum_{p \in \mathbb{Z}} e^{-X_1 p^2} e^{2ip\zeta},$$

whose triple product [24, 38] is

$$\theta_3(\zeta, e^{-X_1}) = \prod_{m \geq 1} (1 - e^{-2mX_1}) \left| 1 + e^{-(2m-1)X_1} e^{2i\zeta} \right|^2, \quad (64)$$

with the first factor independent of ζ . Expanding the logarithm of the second factor at $\zeta = \pi n(\mu, B)$, $2 \operatorname{Re} \ln(1+y)$ with $\ln(1+y) = -\sum_{p \geq 1} (-y)^p/p$, the real part supplies $\cos(2\pi p n(\mu, B))$. At each p the sum over m is geometric, $\sum_{m \geq 1} e^{-(2m-1)X_p} = (2 \sinh X_p)^{-1}$, and

$$-T \ln Z_{\text{zero}} = T \sum_{p \geq 1} \frac{(-1)^p}{p \sinh X_p} \cos(2\pi p n(\mu, B)), \quad (65)$$

up to smooth terms, the Gaussian prefactor of Eq. (61) and the first factor of Eq. (64), which join the nonoscillatory thermodynamics. The alternating sign is absorbed into the phase, $(-1)^p \cos(2\pi p n) = \cos[2\pi p(n + \frac{1}{2})]$. The half-integer offset of the ladder is the origin of the Maslov one-half, here a property of the dual series rather than a quantization input. By Eq. (56) the integer $2\pi/B + C$ drops from every harmonic, the cosine is even, and by Eq. (55) the pocket term of Eq. (56) splits into the $1/B$ frequency term $A_{\mathcal{M}}/2\pi B$ and an $O(1)$ shift. The free energy assembles into

$$\delta\Xi_{\text{osc}} = \frac{B \hbar\tilde{\omega}_{\text{eff}}}{4\pi^3} \sum_{p \geq 1} \frac{X_p}{p^2 \sinh X_p} \cos\left[2\pi p \left(\frac{A_{\mathcal{M}}}{2\pi B} - \tilde{\gamma}\right)\right], \quad (66)$$

the LK form, with the oscillation frequency $A_{\mathcal{M}}/2\pi$ read off Eq. (56) as the pocket area over 2π , and the quantization phase collecting the three $O(1)$ pieces of the count,

$$\tilde{\gamma} = \underbrace{\frac{1}{2}}_{\text{Maslov}} - \underbrace{\frac{1}{2\pi} \int_P \Omega d^2 k}_{\tilde{\gamma}_1 \text{ (Berry)}} - \underbrace{\frac{1}{4\pi} \oint_{\mathcal{C}_0} \frac{\mathcal{X}}{|\nabla \mathcal{M}|} d\ell}_{\tilde{\gamma}_2 \text{ (\mathcal{X} term)}}. \quad (67)$$

Equations (66)–(67) are the central result of this paper: a closed Lifshitz–Kosevich formula for a band with no zero-field Fermi surface, every entry built from the zero-field wavefunctions. Each term has its own origin. The one-half is the half-integer offset of the dual series. The Berry term $\tilde{\gamma}_1$ comes from the measure, the Berry flux accumulated over the pocket interior. The $\mathcal{X}$ term $\tilde{\gamma}_2$ is the first-order shift of the quantized contour produced by the $O(B^2)$ dispersion, surviving as an $O(1)$ phase precisely because of the quadratic spacing (58). Its analog in an ordinary metal is the Roth correction [15], small and usually neglected. The prefactor $B \hbar\tilde{\omega}_{\text{eff}}/4\pi^3$ falls as B^3 , one power from the level degeneracy $B/2\pi$ and

two from the spacing $\hbar\tilde{\omega}_{\text{eff}}$. A metallic spacing, linear in B , gives B^2 . Equation (67) is written for the hole-like pocket used above. For an electron-like contour the count needs no zone term. It is Eq. (54) as it stands, with the flux integral over the occupied pocket P . The harmonics of Eq. (65) are even in n , so the hole-like series may equally be generated by $-n$ of Eq. (56). Comparing $-n$ with Eq. (54) term by term, the leading area term and the Berry term agree, the integer drops from the harmonics as before, and the shift of Eq. (55) enters once directly and once through the complement $4\pi^2 - A_{\text{occ}}(B)$, with opposite sign. Every step from Eq. (65) to Eq. (67) then repeats verbatim, and only the $\mathcal{X}$ term of the phase changes sign:

$$\tilde{\gamma}_2 = \pm \frac{1}{4\pi} \oint_C \frac{\mathcal{X}}{|\nabla\mathcal{M}|} d\ell, \quad (68)$$

the upper sign for an electron-like contour and the lower, the case of Eq. (67), for a hole-like one. In both cases P in $\tilde{\gamma}_1$ is the region the contour bounds (occupied for an electron-like contour, unoccupied for a hole-like one). The hole-like case is the one tested numerically in Sec. VD.

V. NUMERICAL CHECKS

The analytic results above are parameter-free. Every predicted number is computed from zero-field wavefunctions. This section checks them against exact diagonalization (ED) of a lattice model.

A. Lattice model and numerical prescription

We take the flattened two-band model

$$\hat{H}(\mathbf{k}) = \hat{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}, \quad \hat{d} = \frac{\mathbf{d}}{|\mathbf{d}|}, \quad (69)$$

with $\mathbf{d} = (\sin k_x, \sin k_y, u + \cos k_x + \cos k_y)$ the Qi–Wu–Zhang vector [39]. Normalizing $\mathbf{d}$ flattens both bands exactly: $\varepsilon_{\pm} = \pm 1$, full gap $\Delta = 2$, leaving all $\mathbf{k}$ -dependence in the wavefunctions. We study the lower band at $u = -1.2$. The band is isolated for $u \neq 0, \pm 2$. The two bands are related by $\varepsilon \rightarrow -\varepsilon$, so the lower band has $\mathcal{M}(\mathbf{k}) = \Omega(\mathbf{k})$ exactly, a convenience of the two-band limit, not an input to the theory. Figure 1 shows the moment over the zone; it is negative at Γ and X , positive inside a single closed pocket around M bounded by the zero contour C_0 . The pocket area $A_{\mathcal{M}}$, the area enclosed by C_0 , and the contour integral $\oint_{C_0} d\ell/|\nabla\mathcal{M}|$, which set the frequency and the level spacing of the LK formula respectively, are evaluated by contour integration on dense momentum grids, extrapolated to zero grid spacing.

The two-band corollary (35) applies to this model and is used repeatedly below. The ED prescription used for all figures: a rational flux $B = 2\pi/q$ enters through

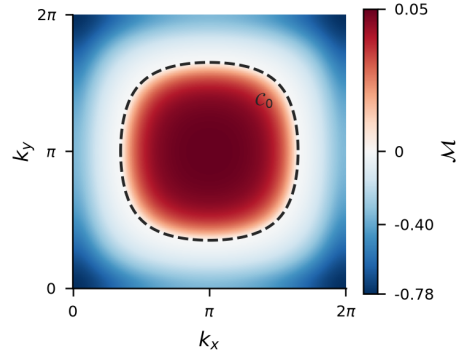


FIG. 1. The orbital moment $\mathcal{M}(\mathbf{k})$ of the lower band at $u = -1.2$, in the zone chart centered on $M = (\pi, \pi)$. The moment is negative at Γ and X , with minimum -0.78 at Γ , and positive inside a single pocket around M , with maximum $+0.05$. The color scale is linear on each side of zero. Dashed: the zero contour C_0 , the flat-band Fermi surface of Sec. IV.

straight-line Peierls substitution on the real-space hoppings of the flattened model, truncated at a hopping range where the exponential tails fall below 10^{-12} per element ($|\mathbf{r}| \leq 30$ at $u = -1.2$). The resulting $2q \times 2q$ Harper matrices are diagonalized exactly for q up to 2560, with Hermiticity residuals below 10^{-11} . The q levels that evolve from the flat band, separated from the rest of the spectrum by the gap, are the flat-band Landau cluster of the checks below.

B. The field-generated spectrum

Figure 2 shows the two structural scalings. The ED bandwidth follows $(\mathcal{M}^{\text{max}} - \mathcal{M}^{\text{min}})B$, and the level spacing at $\mu = \varepsilon_0$ follows the quadratic law of Eq. (58) with the independently computed contour integral $\oint_{C_0} d\ell/|\nabla\mathcal{M}| = 110.65$, both without adjustable parameters. The spacing law involves only $\oint d\ell/|\nabla\mathcal{M}|$ and is therefore independent of $\mathcal{X}$.

C. The smooth response

The smooth part of the ED grand potential is fitted by a quartic polynomial in B over $q \in [320, 1200]$ at $T = 0.1$, where the dHvA terms are thermally negligible, and the linear and quadratic coefficients are compared with $-M$ [Eq. (32)] and $-\chi_{\text{thermo}}/2$ [Eq. (36)]. Figure 3 shows the μ -scan at $T = 0.1$. The ED tracks the closed form through the Curie peak at the band and through the in-gap diamagnetic plateau ($\chi_{\text{thermo}} = -I$ to 0.1% at midgap, with $I = 0.0732$ at $u = -1.2$).

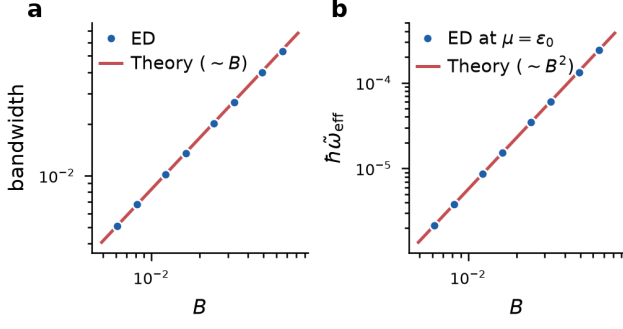


FIG. 2. Log-log plots for the field-generated band at $u = -1.2$. (a) ED bandwidth of the flat-band Landau cluster vs B (points) against $(\mathcal{M}^{\text{max}} - \mathcal{M}^{\text{min}})B$ (line, no fit). (b) ED level spacing at $\mu = \varepsilon_0$ vs B (points) against $\hbar\tilde{\omega}_{\text{eff}} = 2\pi B^2 / \oint_{C_0} d\ell / |\nabla \mathcal{M}|$ [Eq. (58); line, no fit].

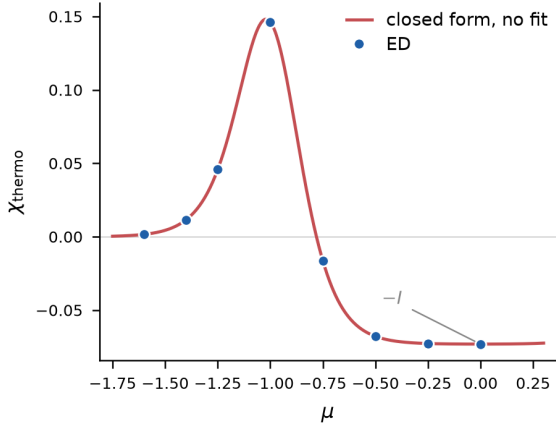


FIG. 3. Orbital magnetic susceptibility of the flat-band model at $u = -1.2$, $T = 0.1$: ED (points; quadratic coefficient of the smooth grand potential, $q = 320$ – 1200) against the closed form Eq. (36) (line, no fit). The paramagnetic peak at $\mu = \varepsilon_0$ is the orbital Curie response of the field-generated moments. The plateau in the gap is the geometric diamagnetism $\chi_{\text{thermo}} = -I$, with no Fermi surface.

D. The dHvA effect with μ pinned to the flat band

Figure 4 shows the LK series against ED; both the waveform against $1/B$ at pinned $\mu = \varepsilon_0$, with frequency $A_{\mathcal{M}}/2\pi = 2.26$, and the fundamental amplitude against $X/\sinh X$ damping on an in-band contour, both with every parameter fixed by geometry. The quantization phase is tested by the same waveform. The curve of Fig. 4(a) uses the zero-parameter prediction $\tilde{\gamma} = 0.874$ of Eq. (67), with pieces $\tilde{\gamma}_1 = -0.078$ and $\tilde{\gamma}_2 = +0.452$. Without the $\mathcal{X}$ term, the phase would be 0.422, nearly half a period away. On this pocket the $\mathcal{X}$ term is the larger, by a factor of 5.8.

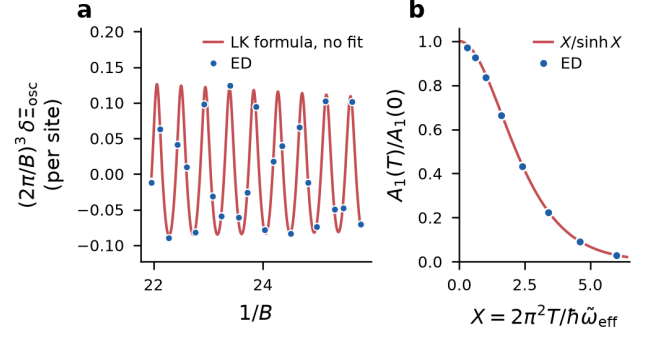


FIG. 4. dHvA oscillations at $u = -1.2$. (a) Oscillatory grand potential vs $1/B$ at pinned $\mu = \varepsilon_0$ and $T = 5 \times 10^{-6}$: ED points after subtracting a polynomial fit of the smooth part in B , rescaled by $(2\pi/B)^3$ to cancel the B^3 envelope of Eq. (66), against the LK formula with all parameters fixed by geometry and $\tilde{\gamma} = 0.874$ (line, no fit). (b) Fundamental amplitude A_1 vs T on the in-band contour $\mathcal{M} = -\frac{1}{2}$ ($\mu = \varepsilon_0 - B/2$, $q = 150$), normalized to $T \rightarrow 0$, against $X/\sinh X$ with $X = 2\pi^2 T / \hbar\tilde{\omega}_{\text{eff}}$ and $\hbar\tilde{\omega}_{\text{eff}}$ from Eq. (58) evaluated on that contour.

VI. A NARROW BAND: CROSSOVER INTO THE FLAT-BAND REGIME

Any flat band realized in a material retains a small dispersion, and the oscillations of Sec. IV apply where the field-generated dispersion dominates it. At weak field the oscillations are the ordinary dHvA of the narrow band. Between the two limits lies a crossover, and its profile follows from the same construction with no new input. Let the band carry a small built-in dispersion, $E_0(\mathbf{k}) = \varepsilon_0 + W g(\mathbf{k})$ with g of order unity and bandwidth $W \ll \Delta$. The field expansion of Sec. II is organized by $x = B\xi^2$, a wavefunction property independent of W , and it is exact at this order when $W = 0$, so every term coupling the built-in dispersion to the field carries at least one power of W . At the retained order the two dispersions add:

$$\varepsilon_W(\mathbf{k}; B) = W g(\mathbf{k}) + B \mathcal{M}(\mathbf{k}) + \frac{1}{2} \mathcal{X}(\mathbf{k}) B^2 + O(Wx, \Delta x^3), \quad (70)$$

measured from ε_0 , with $\mathcal{M}$ and $\mathcal{X}$ the flat-band functionals of Eqs. (7) and (23). The terms by which the second-order energy of the composite band differs from $\frac{1}{2} \mathcal{X} B^2$ all carry W and sit in the $O(Wx)$ error. The built-in term is field independent while the geometric term grows linearly in B , and the larger of the two changes at the crossover field

$$B_{\mathcal{M}} = W \frac{\max_P g - \min_P g}{\max_P \mathcal{M} - \min_P \mathcal{M}}, \quad (71)$$

where both ranges are taken over the occupied region at the filling considered, at weak field the zero-field Fermi sea. Landau quantization proceeds on the *composite* contours. The construction of Sec. IV quantizes the level sets of the first-order dispersion by the area rule, the second-order term entering only through the phase $\tilde{\gamma}_2$ of Eq. (68).

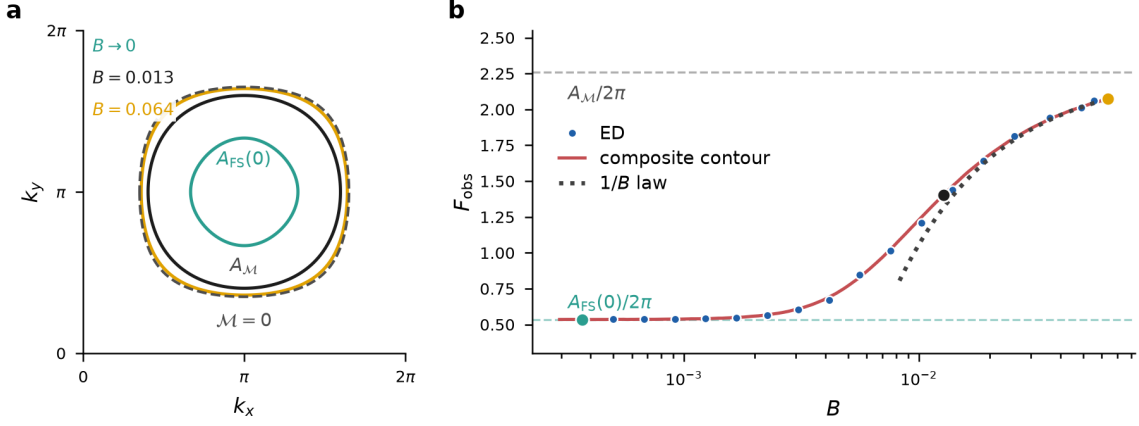


FIG. 5. The dHvA crossover of a narrow band added to the model of Sec. V ($u = -1.2$, $W = 2.8 \times 10^{-4}$, $\mu - \varepsilon_0 = 3.5W$, half a W below the band top; the whole sweep sits inside the expansion window, $x = B\xi^2 \leq 0.10$ with $\xi^2 = 1.6$). (a) The quantized contour $Wg(\mathbf{k}) + B\mathcal{M}(\mathbf{k}) = \mu - \varepsilon_0$ in the zone: at $B \rightarrow 0$ the zero-field Fermi surface, a hole pocket around M with area $A_{FS}(0)$. With increasing field it inflates onto the $\mathcal{M} = 0$ contour (dashed, area A_M), the flat-band Fermi surface of Sec. IV. (b) The frequency it produces: instantaneous frequency, the windowed slope of the level count $N(\mu; q)$ over 25–75 contiguous flux values (points; counts by matrix inertia for q up to 1.7×10^4), against the composite-contour prediction of Eq. (73) (line, no fit), rising from the plateau at $A_{FS}(0)/2\pi$ toward the asymptote $A_M/2\pi$ (dashed) along the $1/B$ law of Eq. (75) (dotted). Colored dots in (b) mark the fields of the contours in (a), the chirp midpoint and the sweep edge.

It applies verbatim with $Wg + B\mathcal{M}$ in place of $B\mathcal{M}$; the same area rule and quantization phase, now on contours of the composite first-order dispersion rather than of $\mathcal{M}$ alone.

The observable consequence is a frequency chirp. The oscillations stay locally periodic in $1/B$ while their frequency drifts across the sweep. At fixed $\mu = \varepsilon_0 + Wg_0$ the quantized contour is the level set $Wg + B\mathcal{M} = Wg_0$. The pocket it bounds, here the unoccupied region, has area $A_{FS}(B) = 4\pi^2 - A_{occ}(B)$, and by Eq. (56) it sets the phase of every harmonic of Eq. (66), the integer zone term $2\pi/B + C$ dropping as in Sec. IV D. A measurement sweeping $1/B$ reads the local frequency

$$F_{obs}(B) = \frac{d}{d(1/B)} \left[\frac{A_{FS}(B)}{2\pi B} \right], \quad (72)$$

and carrying out the derivative gives

$$F_{obs}(B) = \frac{1}{2\pi} \left[A_{FS}(B) - B \frac{dA_{FS}}{dB} \right], \quad (73)$$

where the derivative term is the contribution of the field-dependent area. Where the field-generated term shifts the pocket area appreciably, the derivative term is comparable, and the measured frequency separates from the instantaneous $A_{FS}(B)/2\pi$. The quantization phase $\tilde{\gamma}$ of the composite contour also drifts with B . It contributes $B^2 d\tilde{\gamma}/dB$ to the measured frequency, its accumulated phase over the sweep is bounded by the $O(1)$ range of $\tilde{\gamma}$, and it is negligible against the $1/B$ frequency terms throughout. The two limits of Eq. (73) follow from perturbing the contour about its two endpoints. For $B \ll B_M$ the contour is the zero-field Fermi surface of

the narrow band, and the pocket area drifts linearly in the field-generated term,

$$\begin{aligned} A_{FS}(B) &= A_{FS}(0) + \frac{B}{W} \oint_{FS} \frac{\mathcal{M}}{|\nabla g|} d\ell, \\ F_{obs}(B) &= \frac{A_{FS}(0)}{2\pi} + O(B^2), \end{aligned} \quad (74)$$

the drift cancelling between the two terms of Eq. (73); this is ordinary dHvA, with the zero-field frequency protected to linear order. For $B \gg B_M$ the contour condition becomes $\mathcal{M} = (W/B)(g_0 - g) \rightarrow 0$. The contour flows to $\mathcal{C}_0$, and the oscillations become the flat-band oscillations of Sec. IV, frequency and phase included. The approach to the asymptote is also predicted. Perturbing the contour off $\mathcal{C}_0$ gives

$$\begin{aligned} A_{FS}(B) &= A_M - \frac{W}{B} \Lambda_g, \\ F_{obs}(B) &= \frac{1}{2\pi} \left[A_M - \frac{2\Lambda_g W}{B} \right], \end{aligned} \quad (75)$$

where $\Lambda_g = \oint_{\mathcal{C}_0} (g_0 - g) d\ell / |\nabla \mathcal{M}|$ and A_M is the area enclosed by $\mathcal{C}_0$. The tail's coefficient is a zero-field contour integral, and its factor of two relative to the area law is an artifact of the derivative term in Eq. (73). The same composite-contour rule fixes the other observables across the crossover. The level spacing interpolates between the linear-in- B law of a dispersive band and the quadratic law (58), and the quantization phase interpolates between its endpoint values on the two contour families.

The quantization phase migrates with the contour, but the phase a measurement reads has its own scale, parametrically later than B_M . Inserting Eq. (75) into the

phase of each harmonic of Eq. (66) splits it into three terms,

$$2\pi p \left[\frac{A_{\text{FS}}(B)}{2\pi B} - \tilde{\gamma} \right] = p \frac{A_{\mathcal{M}}}{B} - p \frac{W\Lambda_g}{B^2} - 2\pi p \tilde{\gamma}, \quad (76)$$

the frequency term, the dispersion residual, and the geometric offset. The residual and the offset are equal at

$$B_{\tilde{\gamma}} = \left[\frac{W\Lambda_g}{2\pi|\tilde{\gamma}_1 + \tilde{\gamma}_2|} \right]^{1/2}, \quad (77)$$

where $\tilde{\gamma}_{1,2}$ are their $\mathcal{C}_0$ values. The Maslov half, common to every dHvA series, drops from the comparison. Both terms scale as p , so $B_{\tilde{\gamma}}$ is the same for every harmonic. Beyond $B_{\tilde{\gamma}}$, the residual dispersion averaged over $\mathcal{C}_0$ is smaller than the geometric fraction $|\tilde{\gamma}_1 + \tilde{\gamma}_2|$ of the level spacing $\hbar\tilde{\omega}_{\text{eff}}$, and the phase is carried by its Berry and $\mathcal{X}$ terms. The two scales balance the built-in dispersion against the two orders of the field expansion, the bandwidth of the first-order term at $B_{\mathcal{M}}$ and the level spacing $\hbar\tilde{\omega}_{\text{eff}}$ at $B_{\tilde{\gamma}}$, so $B_{\mathcal{M}} \propto W$ while $B_{\tilde{\gamma}} \propto W^{1/2}$. The window $B_{\mathcal{M}} \ll B \ll B_{\tilde{\gamma}}$, with the frequency saturated but the phase still dispersion-dominated, widens as the band flattens.

We verify the crossover by exact diagonalization on the model of Sec. V augmented by a scalar dispersion $Wg(\mathbf{k})\sigma^0$, with σ^0 the identity in orbital space and $g = 2 - \cos k_x - \cos k_y$, chosen because it commutes with the geometric part, so $\mathcal{M}$ and $\mathcal{X}$ are exactly unchanged while the band acquires the bandwidth $4W$. The chemical potential is placed just below the top of the narrow band, and the bandwidth is chosen so that the whole field sweep stays inside the expansion window $x = B\xi^2 \ll 1$ (parameters in the caption of Fig. 5). The zero-field Fermi surface is a hole pocket around M , and with growing field it inflates onto the same $\mathcal{C}_0$ whose oscillations Secs. IV and V quantify, the quantized contour staying hole-like for the whole sweep. Figure 5 shows the measured instantaneous frequency across the sweep. It follows the composite-contour prediction of Eq. (73) with no fitted parameters, from the low-field plateau at $A_{\text{FS}}(0)/2\pi$ through the full chirp. The plateau shows the cancellation of Eq. (74). Deep in the plateau the pocket area has already grown by several tens of percent while the measured frequency has barely moved. The $1/B$ law of Eq. (75) carries the strong-field end, where the measured frequency still sits visibly below the asymptote, by the residual $2\Lambda_g W/B$ at the edge of the sweep. The same sweeps test the spacing interpolation. The measured level-spacing exponent crosses over from 1 in the weak-field window to 2 in the strong-field window, as the composite contour predicts.

The phase scale $B_{\tilde{\gamma}}$ of Eq. (77) lies a factor of two beyond the strong-field edge of the sweep for this protocol, so the frequency sweep does not test the phase. The phase is measured separately, per flux value, from the

level positions, as the fractional index of the level nearest μ within the Landau cluster, referred to the composite pocket area $A_{\text{FS}}(B)$ and differenced against a $W = 0$ sweep on the same flux grid. The measured phase follows Eqs. (67) and (68) evaluated on the composite contour at every field, migrating from the narrow-band value to the $\mathcal{C}_0$ value of Sec. VD, more than a third of a period away, with the midpoint of the migration inside the chirp, and it has essentially reached the $\mathcal{C}_0$ value at $B_{\tilde{\gamma}}$. What changes there is the composition of a phase reading. A constant-frequency fit carries the dispersion residual $W\Lambda_g/2\pi B^2$ of Eq. (76) in its intercept, alongside the geometric offset, and the measured contamination crosses the geometric content at $B_{\tilde{\gamma}}$, as Eq. (77) prescribes.

VII. CONCLUSIONS

In this paper, we have shown that a band with no dispersion has an orbital response determined entirely by wavefunction geometry. The magnetic field generates the dispersion, the moment term $B\mathcal{M}(\mathbf{k})$ at first order and the coefficient $\mathcal{X}(\mathbf{k})$ at second, and the single parameter $x = B\xi^2$ controls the expansion. The susceptibility that follows is purely geometric and consistent with the exact resolvent response formula for any isolated, nondegenerate flat band. Bosonizing the emergent zero contour gives a modified Lifshitz–Kosevich formula, obtained without invoking the semiclassical quantization rule of cyclotron orbits, whose quantization phase carries the second-order geometry at $O(1)$. In a narrow band the dHvA frequency and the quantization phase migrate from their zero-field values to the flat-band ones at the crossover field $B_{\mathcal{M}} \propto W$. The phase content of the signal becomes geometric only at the parametrically later $B_{\tilde{\gamma}} \propto W^{1/2}$, and below that scale the offset a dHvA fit returns is dominated by the dispersion residual rather than by the Berry and $\mathcal{X}$ terms.

Experimentally, the quadratic spacing law sets the observation window. Oscillations require $T \lesssim \hbar\tilde{\omega}_{\text{eff}}/2\pi^2 = B^2/\pi\oint_{\mathcal{C}_0} d\ell/|\nabla\mathcal{M}|$. For a flat band with ξ a few lattice constants ($\oint d\ell/|\nabla\mathcal{M}| \sim 10^2$ in lattice units, as for the model here), a laboratory field of a flux quantum per hundred unit cells ($B \sim 0.06$), and a typical gap of tens of meV as the unit of energy, the window is millikelvin temperatures, standard for dHvA experiments. Disorder is expected to enter as a Dingle factor on the same harmonics [4, 26], to be compared with $\hbar\tilde{\omega}_{\text{eff}}$, the more stringent constraint for flat bands. Of the observables, the magnetization phase is the most demanding. The most accessible geometric channel is the smooth susceptibility plateau, which requires only $T \ll \Delta$.

Flat-band platforms with tunable geometry, moiré systems foremost, are the natural place to look. Interactions, which flat bands amplify, are beyond the present noninteracting treatment. For the smooth response a concrete interacting bridge exists through mean-field bands [21], while an interacting theory of the oscillatory

response remains open.

Appendix A: Consistency with the gauge-invariant susceptibility

This appendix reduces the exact trace formula of Ref. [18], restated as Eq. (34) (the noninteracting limit of the interacting formulation of Ref. [21]), at an arbitrary isolated, nondegenerate flat band of an N -band Hamiltonian, and shows that it reproduces the structure and the coefficients of Eq. (33).

Split the resolvent at the flat energy ($P = |v^0\rangle\langle v^0|$ the flat-band projector, $Q = 1 - P$):

$$G(E) = \frac{P}{E - \varepsilon_0 + i0} + R(E), \quad R(E) \equiv Q(E - h)^{-1}Q, \quad (\text{A1})$$

with $R(E)$ analytic inside the flat band's spectral window, the interval of half-width Δ around ε_0 that contains no other spectrum. We write $R_0 \equiv R(\varepsilon_0)$.

Differentiating the projector form of the eigenvalue equation, $hP = \varepsilon_0 P$, gives $h^a P + h \partial_a P = \varepsilon_0 \partial_a P$. Multiply by P from both sides:

$$P h^a P + P h \partial_a P P = \varepsilon_0 P \partial_a P P. \quad (\text{A2})$$

Since $Ph = \varepsilon_0 P$, the second term on the left equals the right-hand side, and the two cancel,

$$P h^a P = 0. \quad (\text{A3})$$

A flat band carries no velocity matrix element.

Now insert Eq. (A1) into the trace of Eq. (34). Each of the four resolvent slots contributes either the pole term $P/(E - \varepsilon_0 + i0)$ or the analytic term $R(E)$, so the trace organizes by powers of the pole factor,

$$\text{Tr}\{\dots\} = \frac{\kappa_2(\mathbf{k})}{(E - \varepsilon_0 + i0)^2} + \frac{\kappa_1(\mathbf{k})}{E - \varepsilon_0 + i0} + (\text{analytic}), \quad (\text{A4})$$

and this display defines $\kappa_{1,2}$. Here κ_2 collects the placements with two P 's and the remaining slots at R_0 . The coefficient κ_1 collects the placements with one P , together with the two- P placements in which one $R(E)$ is replaced by its first Taylor coefficient at ε_0 (the factor $E - \varepsilon_0$ so produced reduces the double pole to a simple one). Higher poles are absent. A pole of order p requires p factors of P , any two P 's in cyclically adjacent slots sandwich a velocity vertex, which vanishes by Eq. (A3), and with four slots any three or four P 's must include an adjacent pair (adjacency is understood cyclically, as in the trace). Poles of order three and four therefore vanish identically.

The energy integral of Eq. (34) is then evaluated in two steps. For the simple pole, $\text{Im}(E - \varepsilon_0 + i0)^{-1} = -\pi \delta(E - \varepsilon_0)$ reduces the integral to $f(\varepsilon_0)$. For the double pole, one integration by parts in E moves the derivative

onto f and reduces the pole order by one, leaving $f'(\varepsilon_0)$. With the $-\frac{1}{12}$ prefactor of Eq. (34),

$$\chi_{\text{thermo}} = \chi_{\text{frozen}} + \frac{f(\varepsilon_0)}{12} \int \frac{d^2 k}{(2\pi)^2} \kappa_1 + \frac{f'(\varepsilon_0)}{12} \int \frac{d^2 k}{(2\pi)^2} \kappa_2, \quad (\text{A5})$$

where χ_{frozen} is the same energy integral with every slot at $R(E)$: the remote bands' contribution, whose μ -dependence inside the window is $O(e^{-\Delta/T})$. No f'' or higher terms appear. The master formula produces *exactly* the affine structure of Eq. (33), for any N .

The two coefficients then match separately, because $f(\varepsilon_0)$ and $f'(\varepsilon_0)$ vary independently with (μ, T) inside the window. For the double pole, write $\mathcal{W}_{ab} \equiv \langle v^0 | h^a R_0 h^b | v^0 \rangle$. The identity $Q |\partial_a v^0\rangle = R_0 h^a |v^0\rangle$, obtained by inverting Eq. (5) on the Q subspace, converts the projected Hessian into this matrix,

$$\langle v^0 | h^{ab} | v^0 \rangle = -2 \text{Re } \mathcal{W}_{ab}, \quad (\text{A6})$$

and $\mathcal{W}_{aa}$ is real. In the two-slot group of Eq. (34) the only two- P placement puts both resolvents at the pole, and Eq. (A6) gives

$$\begin{aligned} & \langle v^0 | h^{xx} | v^0 \rangle \langle v^0 | h^{yy} | v^0 \rangle - \langle v^0 | h^{xy} | v^0 \rangle^2 \\ &= 4 [\text{Re } \mathcal{W}_{xx} \text{Re } \mathcal{W}_{yy} - (\text{Re } \mathcal{W}_{xy})^2]. \end{aligned} \quad (\text{A7})$$

In the four-slot group the P 's occupy the two cyclically non-adjacent slots, the remaining slots carry R_0 , and the two placements of each ordering give

$$\begin{aligned} \text{Tr}\{h^x G h^x G h^y G h^y G\}|_{PP} &= \mathcal{W}_{xx} \mathcal{W}_{yy} + |\mathcal{W}_{xy}|^2, \\ \text{Tr}\{h^x G h^y G h^x G h^y G\}|_{PP} &= \mathcal{W}_{xy}^2 + (\mathcal{W}_{xy}^*)^2. \end{aligned} \quad (\text{A8})$$

Combining Eqs. (A7) and (A8) with the -4 of Eq. (34), and using $|\mathcal{W}_{xy}|^2 - \mathcal{W}_{xy}^2 - (\mathcal{W}_{xy}^*)^2 = 3(\text{Im } \mathcal{W}_{xy})^2 - (\text{Re } \mathcal{W}_{xy})^2$, the determinant combination $\text{Re } \mathcal{W}_{xx} \text{Re } \mathcal{W}_{yy} - (\text{Re } \mathcal{W}_{xy})^2$ cancels between the two groups, leaving

$$\kappa_2 = -12 (\text{Im } \mathcal{W}_{xy})^2 = -12 \mathcal{M}^2, \quad (\text{A9})$$

where the last equality is Eq. (7) reduced by the same Q -subspace identity: $\langle \partial_a v | (\hat{H} - \varepsilon_0) | \partial_b v \rangle = -\langle v | \partial_a \hat{H} R_0 \partial_b \hat{H} | v \rangle = -\mathcal{W}_{ab}$, so $\mathcal{M} = -\text{Im } \mathcal{W}_{xy}$. The Curie coefficient of the master formula is the moment squared, pointwise and for any N . It is governed by the moment, not the curvature (the two coincide only in the two-band case). This reproduces the $\mathcal{M}^2 f'$ term of Eq. (33) identically. For the simple pole the statement is integral rather than pointwise. The residue sums of κ_1 and the dispersion-route coefficient $-\mathcal{X} + 2\mathcal{M}\Omega$ differ pointwise by total derivatives (integration by parts reshuffles a coefficient against the measure without changing any integrated observable), but their zone integrals agree,

$$\int \frac{d^2 k}{(2\pi)^2} [\mathcal{X} + 2\mathcal{M}\Omega] = -\frac{1}{12} \int \frac{d^2 k}{(2\pi)^2} \kappa_1. \quad (\text{A10})$$

Equation (A10) holds because both routes evaluate the same grand potential to $O(B^2)$, Eq. (34) exactly and the dispersion route through the magnetic band construction of Sec. II. We have not reduced the residue sum of κ_1 to the dispersion-route integrand term by term. Numerically we verify Eq. (A10) and the full (μ, T) -dependence of Eq. (A5) on the lattice model of Sec. V at the 10^{-10} level.

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